Lec-17

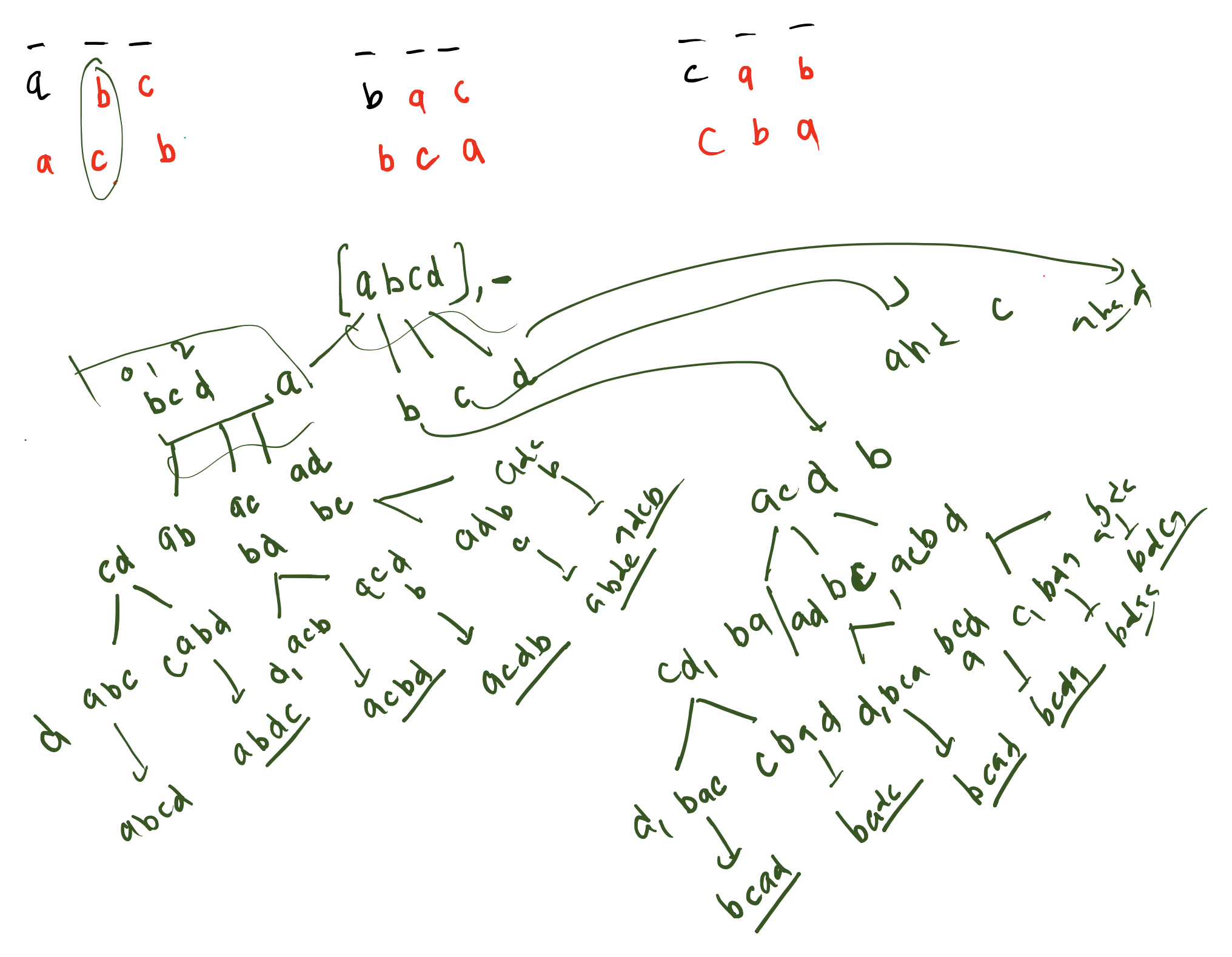
25/04/26 9:51 AM

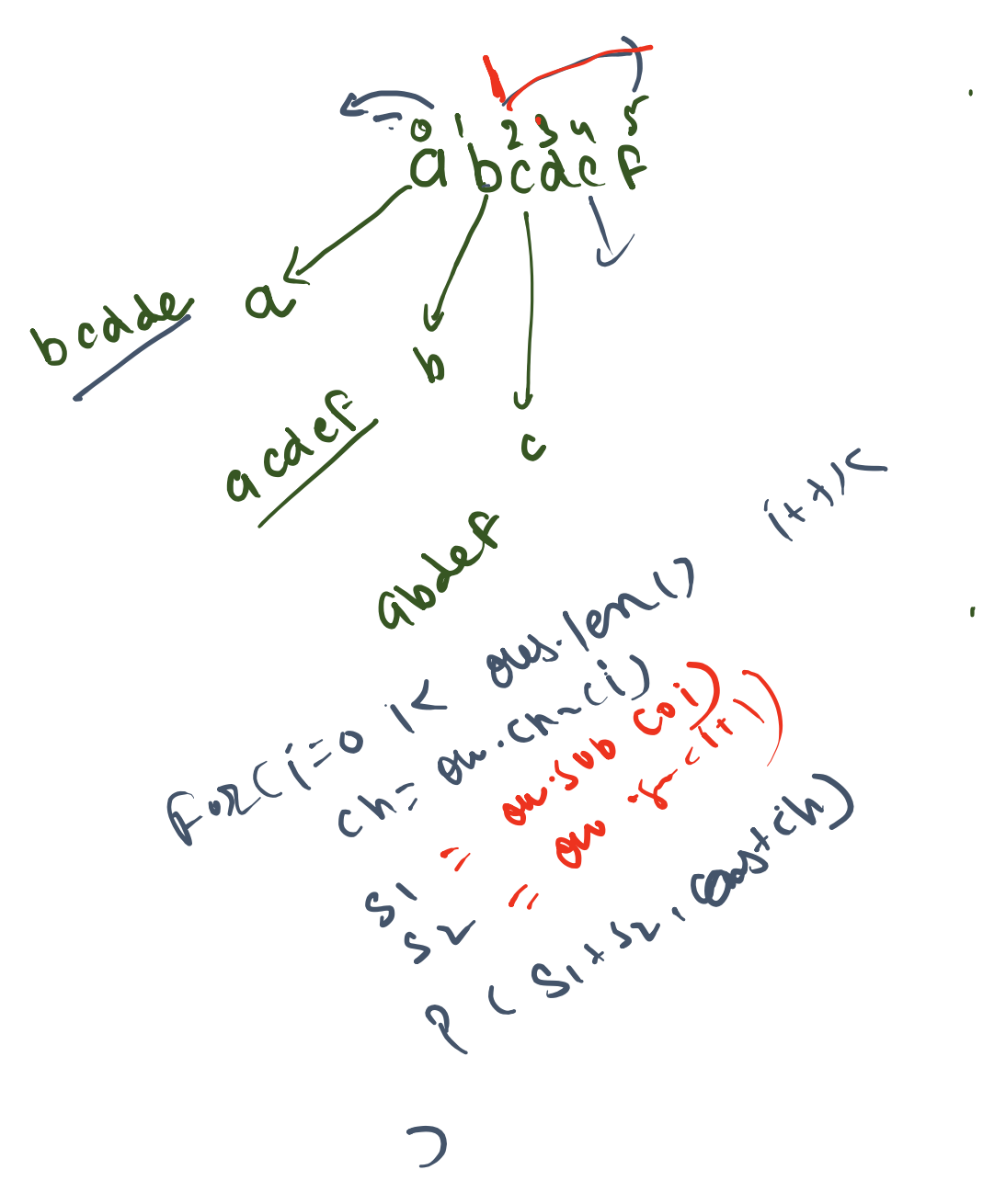
String = abc ↳ permutation

abc ⇒ abc, acb, bac, bca, cab, cba

N → N!

[3 2 1]

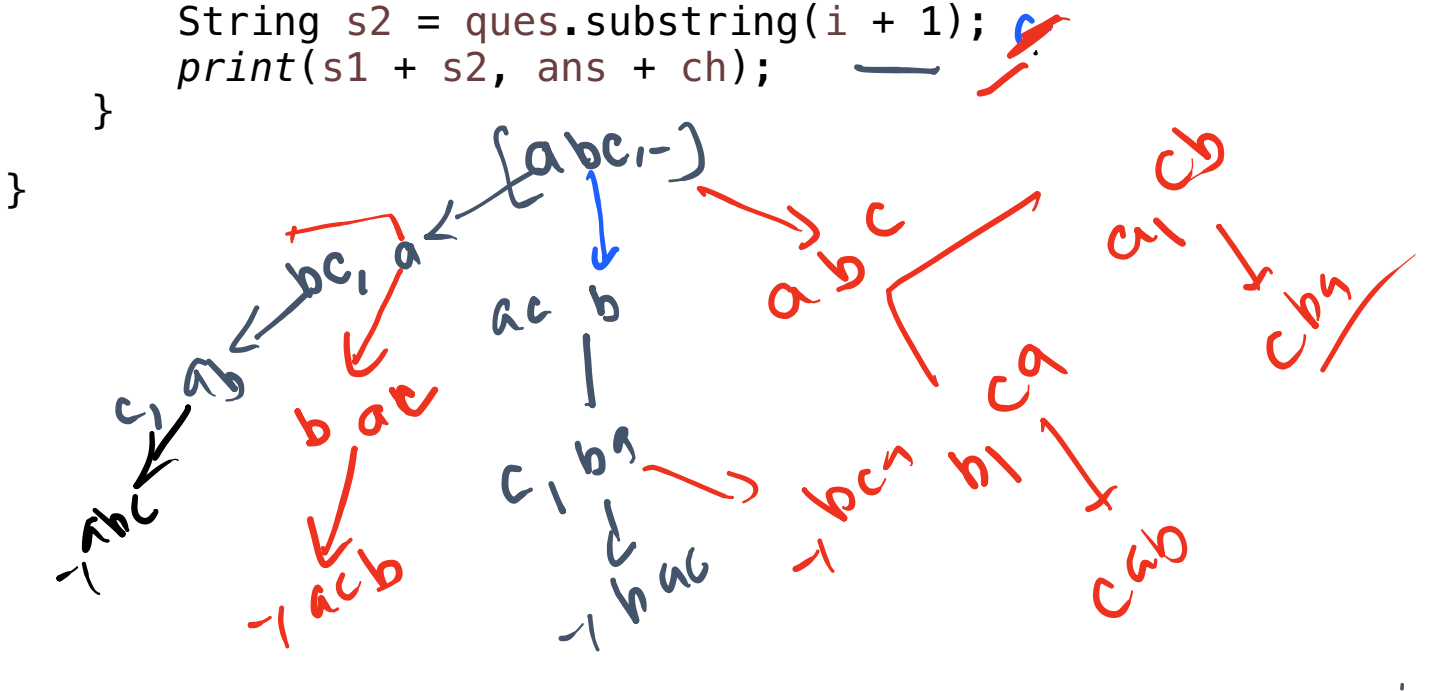

a b c
a c b
b a c
b c a
c a b
c b a
[abcd], -
bcd a
ans c
abcd
acd b
abcd
abdc
acbd
acdb
adbc
adcb
bcad
bdac
bdca

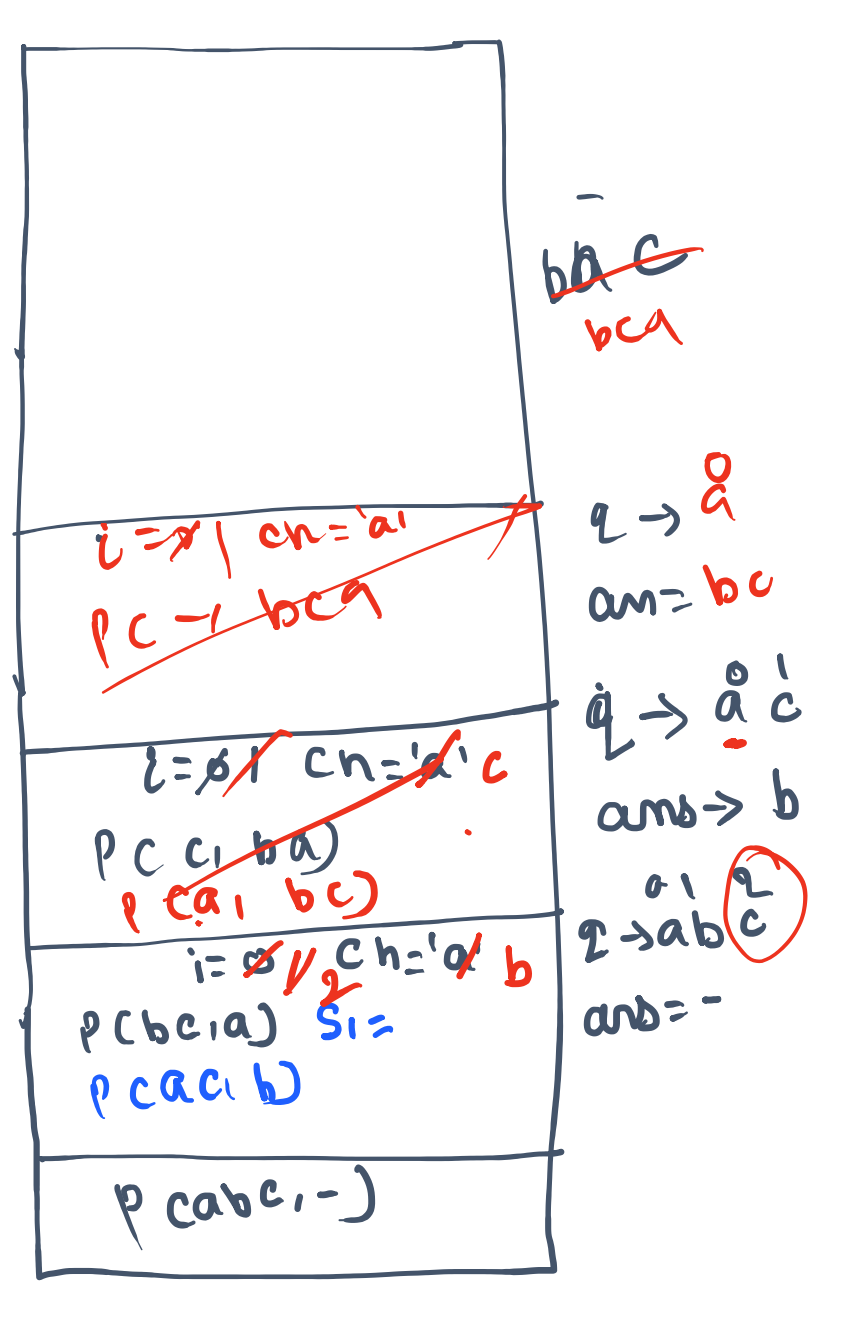

abcdef
bcdef a
acdef b
abdef c
for(i=0 i< ques.len()
ch = ques.charAt(i)
s1 = ques.sub(0,i)
s2 = ques.sub(i+1)
p(s1+s2, ans+ch)

i	s1	s2	s1+s2	ans
i=0	s1 = – (0,0)	s2 = bcdef (1)	s1+s2 bcdef	ans + ch → 'a'
i=1	s1 = a (0,1)	s2 = cdef (2)	s1+s1 acdef	ans + ch → 'b'
i=2	s1 = ab (0,2)	s2 = def (3)	s1+s2 abdef	ans+ch → 'c'
i=3	s1 = abc (0,3)	s2 = ef (4)	s1+s1 abcef	ans+ch → 'd'
i=4	s1 = abcd (0,4)	s2 = f (5)	s1+s1 abcdf	ans+ch → 'e'
i=5	s1 = abcde (0,5)	s2 = -- (6)	s1+s1 abcde	ans = ch → 'f'

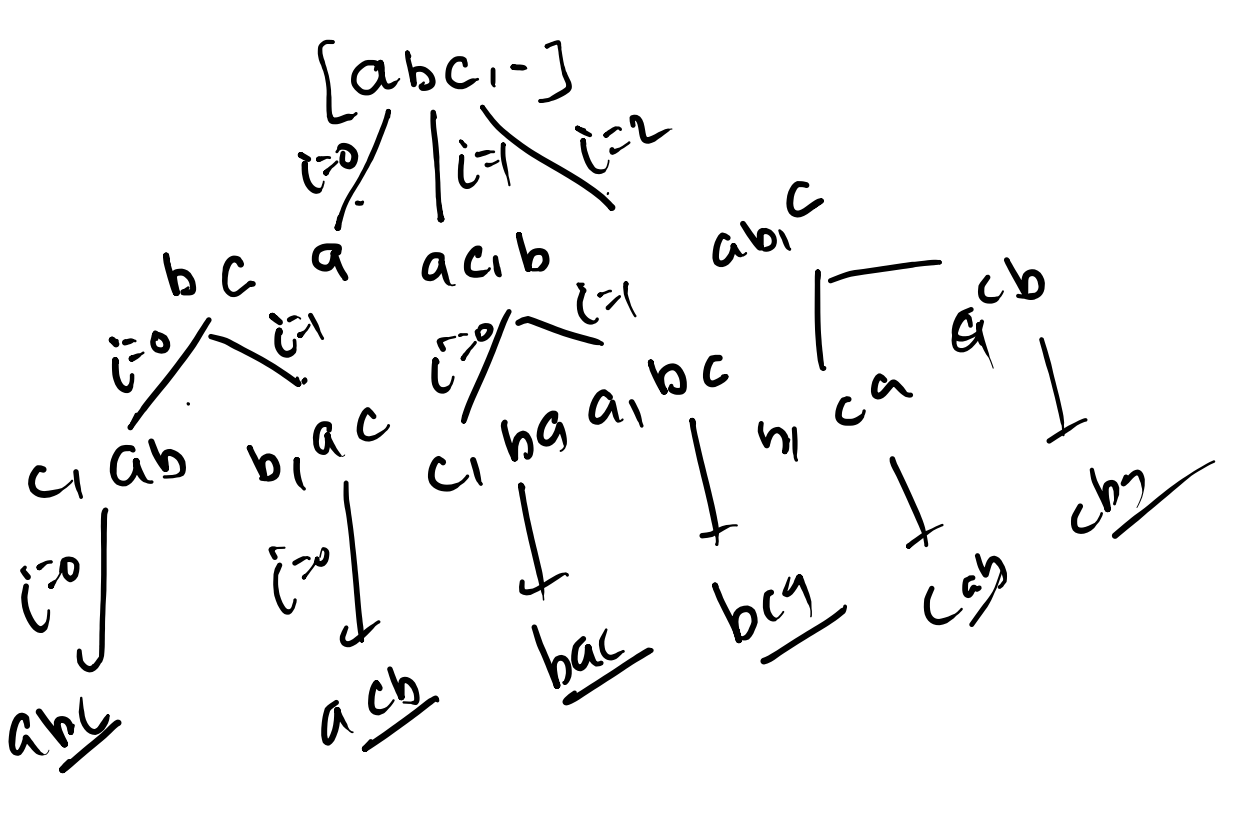
```
public static void print(String ques, String ans) {
    if (ques.length() == 0) {
        System.out.println(ans);
        return;
    }

    for (int i = 0; i < ques.length(); i++) {
        char ch = ques.charAt(i);
        String s1 = ques.substring(0, i);
        String s2 = ques.substring(i + 1);
        print(s1 + s2, ans + ch);
    }
}
```

abc
acb
bac
bca


[abc, -]
bc, a
ac b
ab c
c, ab
b ac
c, ba
bca
cab
abc
acb
bac
cb

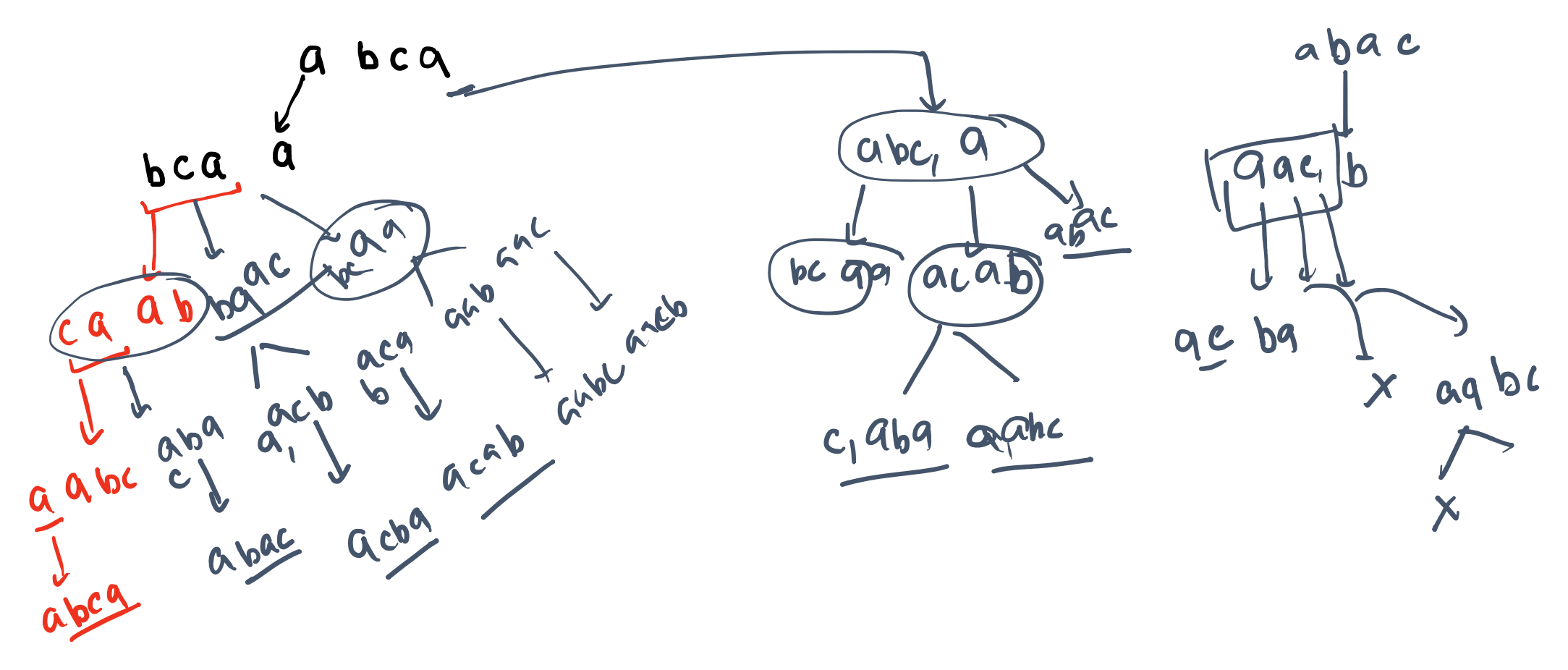

bac
bca
i=1 ch='a'
p(c, bca)
i=0 ch='c'
p(c, ba)
p(ca, bc)
i=0 ch='b'
p(bc, a)
p(ac, b)
main
p(abc, -)
a → a
ans = bc
a → a c
ans → b
a → ab c
ans = -

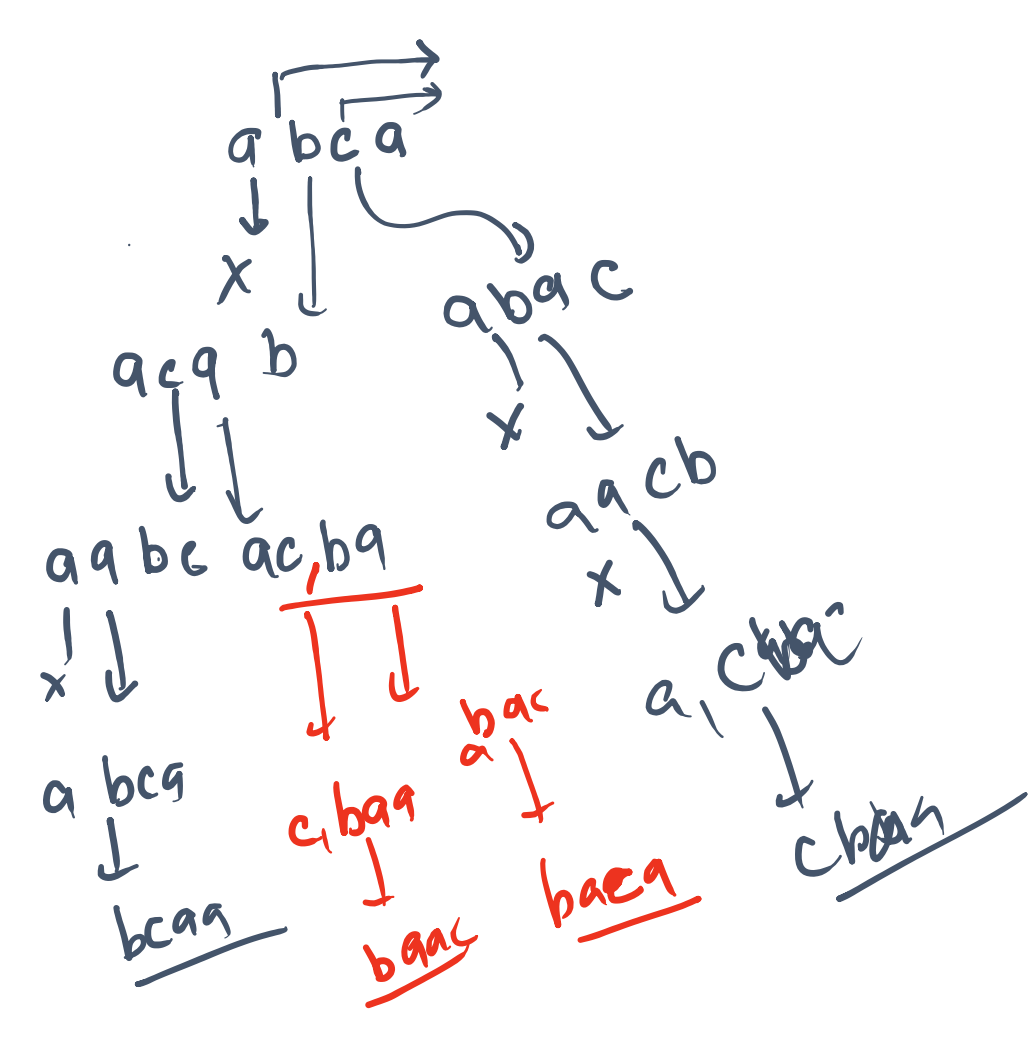
```
public static void print(String ques, String ans) {
    if (ques.length() == 0) {
        System.out.println(ans);
        return;
    }

    for (int i = 0; i < ques.length(); i++) {
        char ch = ques.charAt(i);
        String s1 = ques.substring(0, i);
        String s2 = ques.substring(i + 1);
        print(s1 + s2, ans + ch);
    }
}
```


[abc, -]
i=0 i=1 i=2
bc a
ac, b
ab, c
c, ab
b, ac
c, ba
a, bc
b, ca
a cb
abc
acb
bac
bca
cab
cba

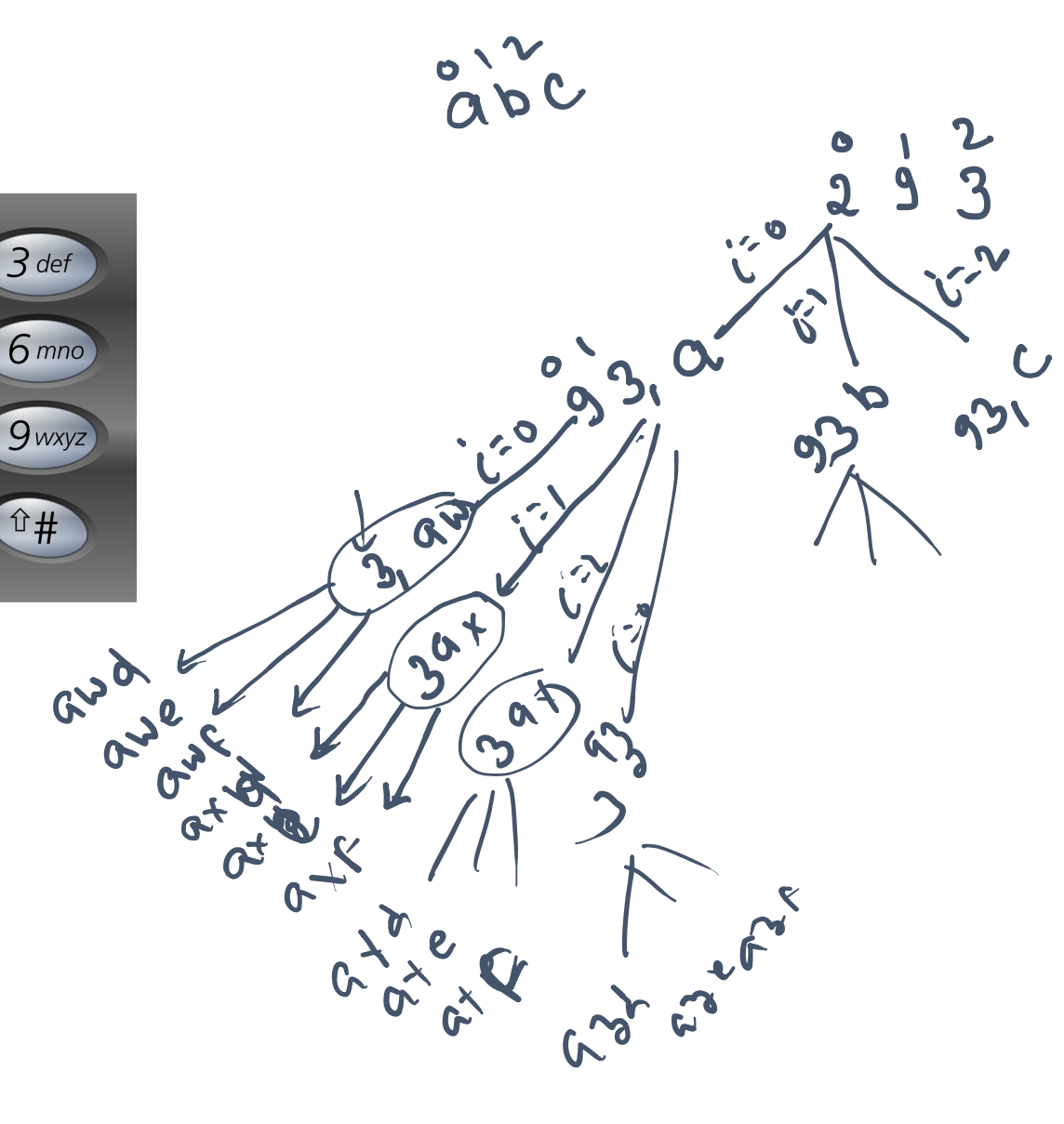
✓ abca
✓ abac
✓ acba
✓ acab
✓ aabc
✓ aacb
✓ baca
✓ baac
✓ bcaa
✗ bcaa
✗ baac
✗ baca

✓ caba
✓ caab
✓ cbaa
✗ cbaa
caab
✗ caba
aabc ✗
aacb ✗
abac ✗
abca ✗
acab ✗
acba ✗


a bca
bca a
ca ab
aabc
abca
abac
acab
abc, a
bc aa
ac ab
c, aba
a, abc
abac
aac b
ac ba
aa bc


abca
aca b
aba c
aa bc
ac ba
aa cb
a bca
c, baa
bac
a, cba
bcaa
baac
baca
cbaa

1
2 abc
3 def
4 ghi
5 jkl
6 mno
7 pqrs
8 tuv
9 wxyz
*
0
#


abc
2 9 3
93, a
93 b
93, c
3, ad
3, ae
3, af
awd
awe
awf
axd
axe
axf
ayd
aye
ayf
azd
aze
azf

232

'2' → 2

ch - 'a' = 0

ch - 'b' = 1

ch - '0' =

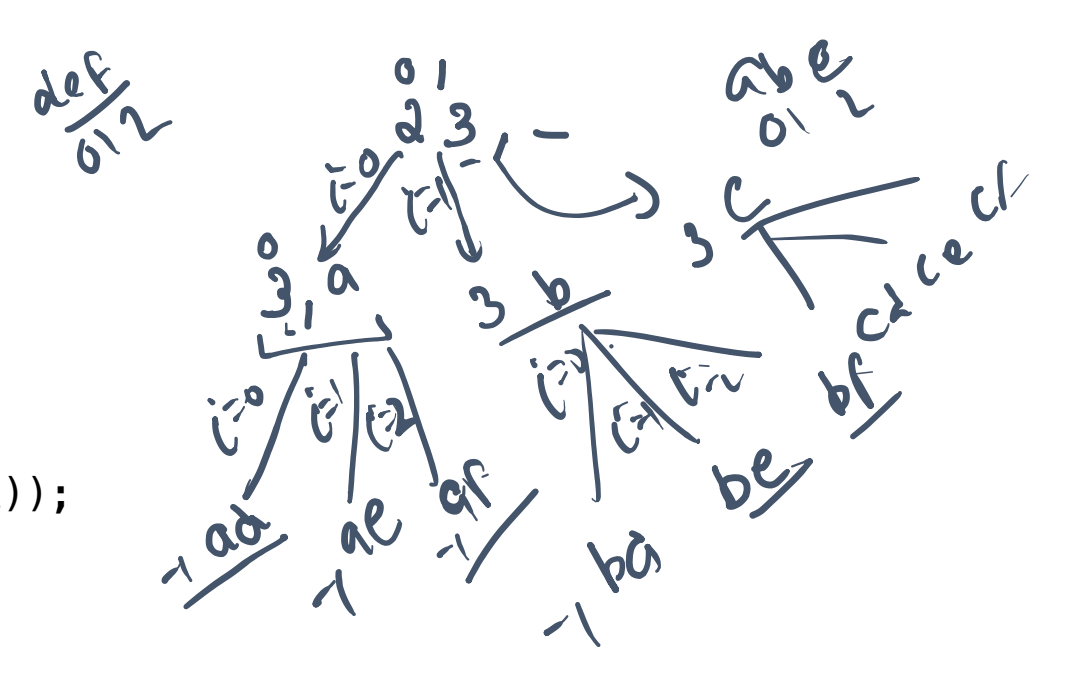
'0' - '0' = 0

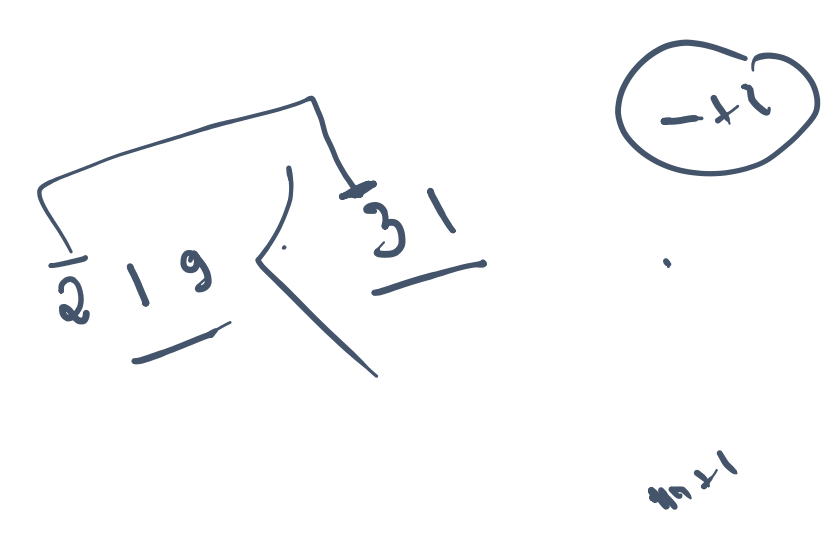
'1' - '0' = 1

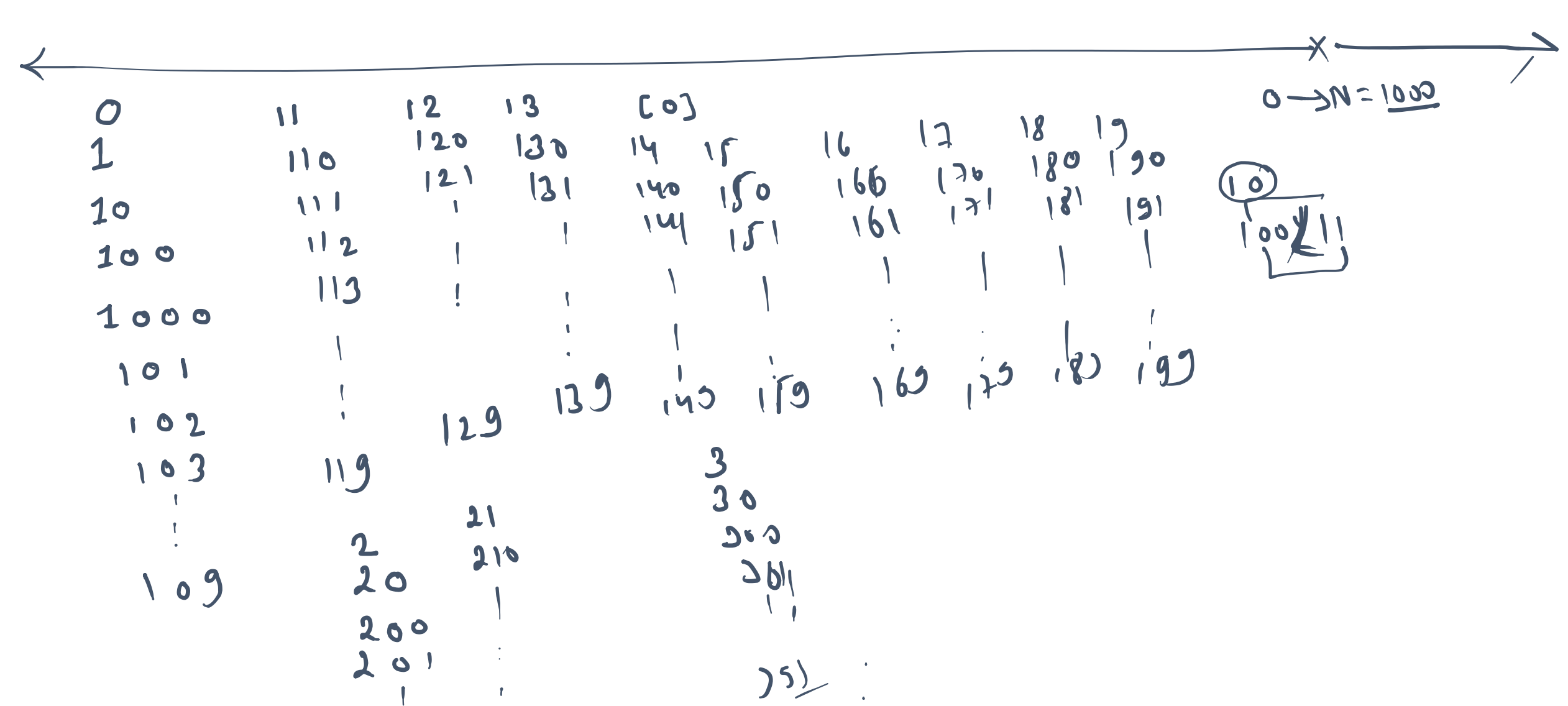
'2' - '0' = 2

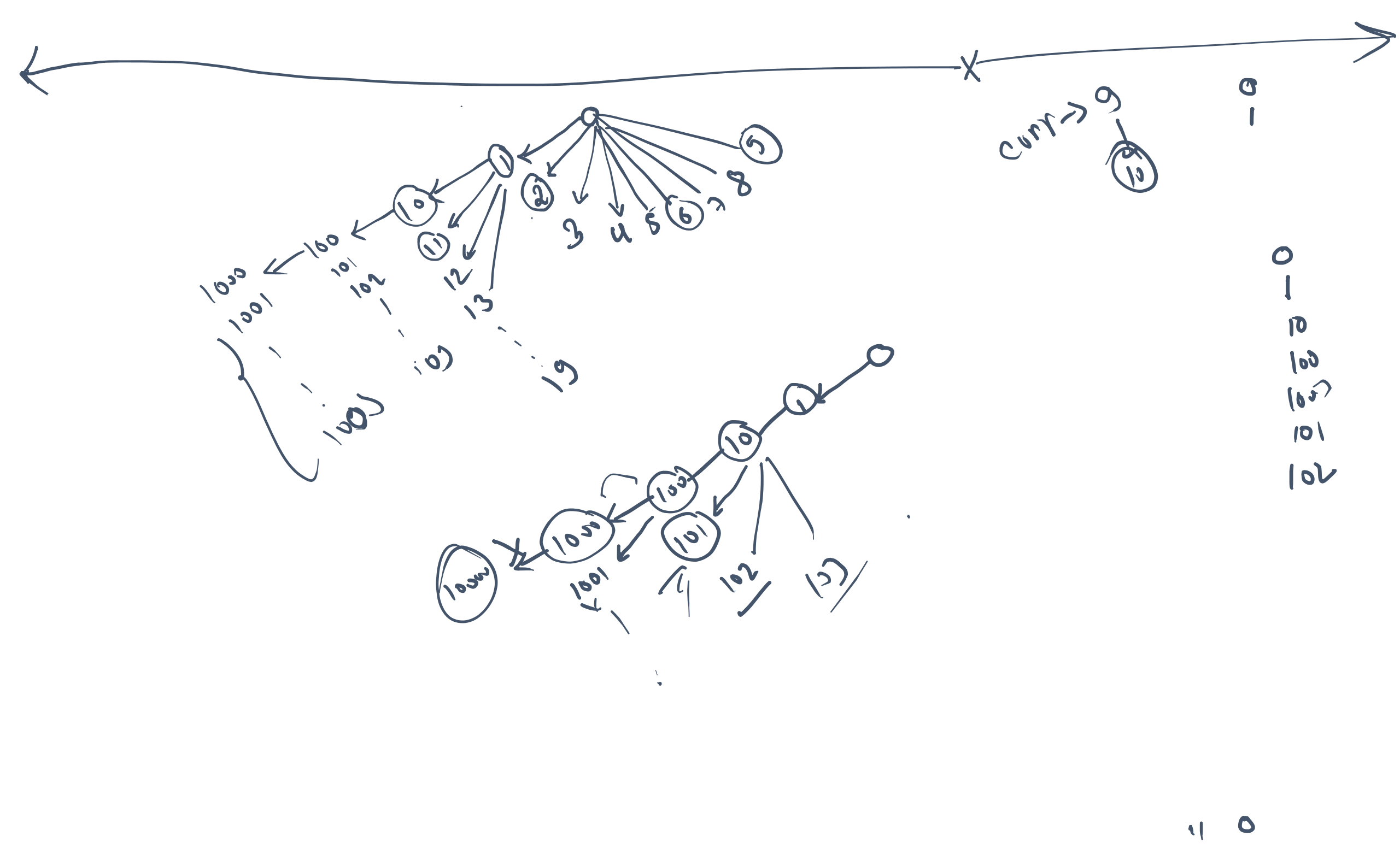
'3' - '0' = 3

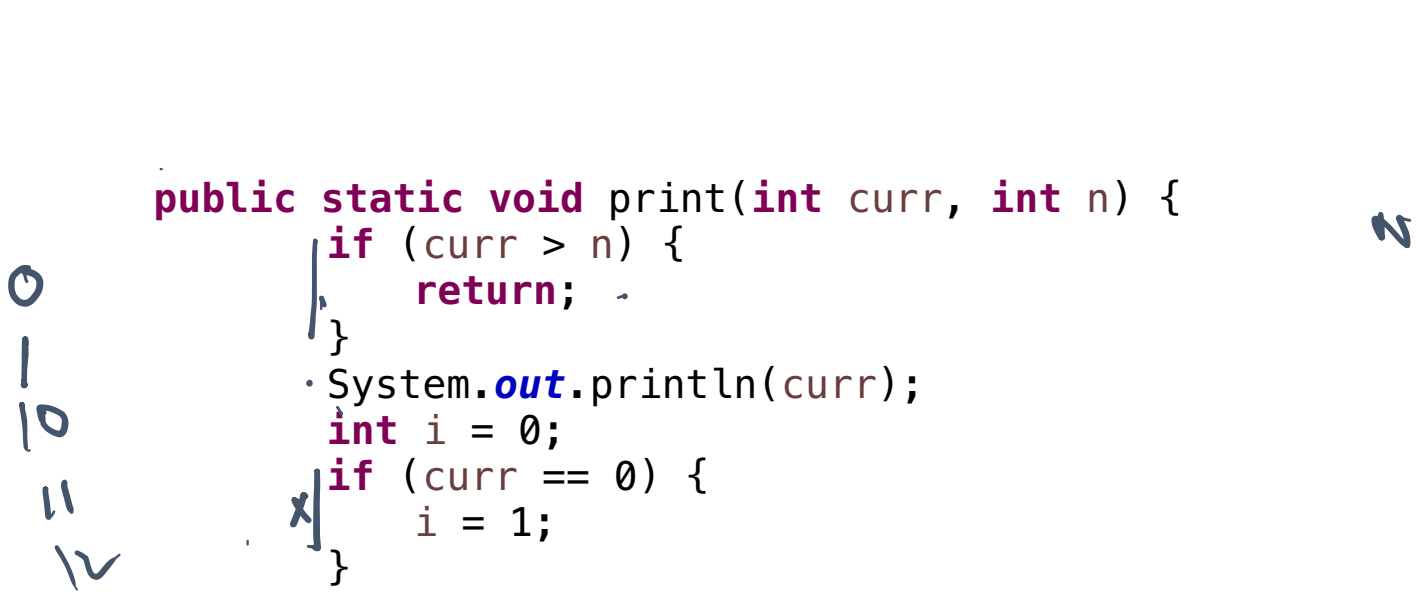
```
public static void KeyPaid(String ques, String ans) {
    if(ques.length()==0) {
        System.out.println(ans);
        return;
    }
    char ch =ques.charAt(0);//'2'
    String str=key[ch-'0'];// abc
    for (int i = 0; i < str.length(); i++) {
        KeyPaid(ques.substring(1), ans+str.charAt(i));
    }
}
```


def
012
abc
012
23
3, a
3 b
3 c
ad
ae
af
bd
be
bf
cd ce cf




0 → N = 1000
0
1
10
100
1000
101
102
103
109
11
110
111
112
113
119
2
20
200
201
209
12
120
121
129
21
210
219 - 299
13
130
131
139
3
30
300
301
351
14
140
141
149
15
150
151
159
16
160
161
169
17
170
171
179
18
180
181
189
19
190
191
199
10
100


curr → 9
10
0
1
10
100
1000
101
102
2
3 4 5 6 7 8 9
11
12
13
19
1000
1001
1009
100
101
102
103
10000


i=1
i=0
1
10
100
1000
1001
101
102
103
11
110

0
1
10
100
1000
101
102
103
!
109

0
1
10
11
12
(2)

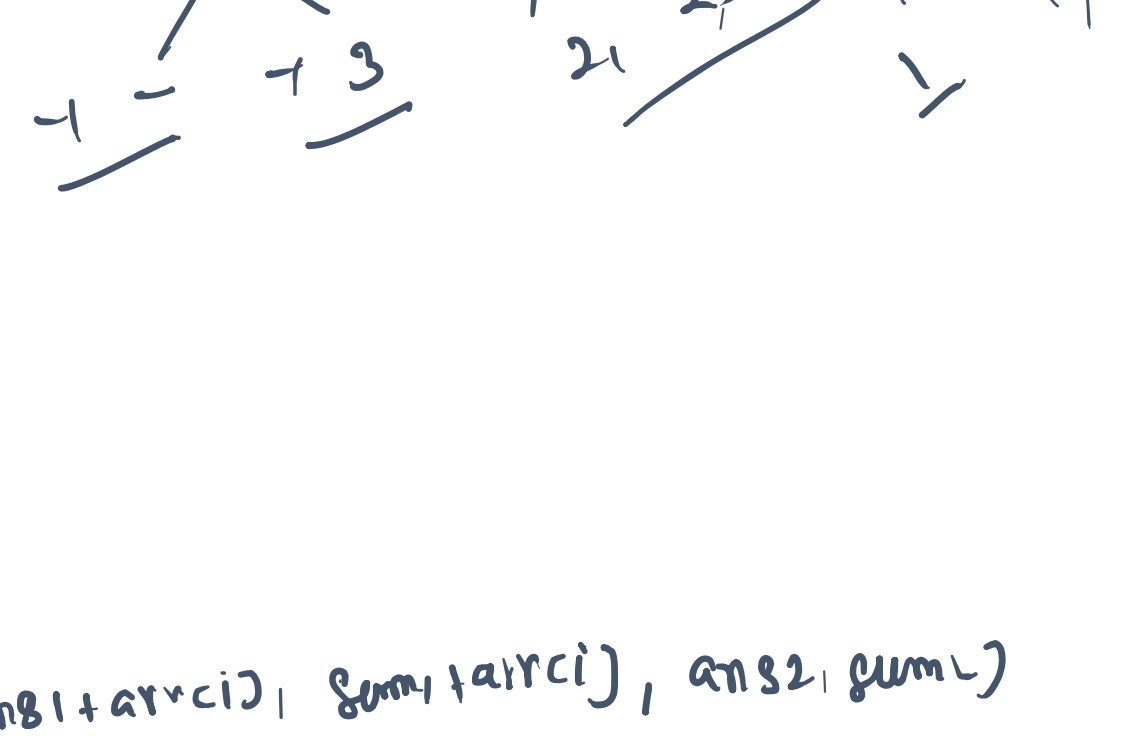
```
public static void print(int curr, int n) {
    if (curr > n) {
        return;
    }
    System.out.println(curr);
    int i = 0;
    if (curr == 0) {
        i = 1;
    }
    for (; i <= 9; i++) {
        print(curr * 10 + i, n);
    }
}
```

0x10+1

1x10+0

10x0

N=12
1
2
9
10
11
12
i=1
i=2


[1, 2, 3]
2, 3
3, 1
3, 2
2, 3
3, 1
3, 1 2
1, 3
1 2
1, 2 3

x = R (curr, i+1, ans1+arr[i], sum1+arr[i], ans2, sum2)

y = R (curr, i+1, ans1, sum1, ans2+arr[i], sum2+arr[i])

if(i==n){
if(sum1==sum2){
ans2
ret 1
}
ret 0
}